competitive game, because when one player is winning the other player is losing; it is zero-sum. Team-based games, like basketball or football, are called *mixed cooperative-competitive games*, since the agents on the same team need to cooperate in order to maximize their rewards, but when one team as a whole is winning, the other team must be losing, so at the team-to-team level it is a competitive game.

In this section we are going to use an open source Gridworld-based game that is specially designed for testing multi-agent reinforcement learning algorithms in cooperative, competitive, or mixed cooperative-competitive scenarios (figure 9.23). In our case, we will set up a mixed cooperative-competitive scenario with two teams of Gridworld agents that can move around in the grid and can also attack other agents on the opposing team. Each agent starts with 1 “health point” (HP), and when they’re attacked the HP decreases little by little until it gets to 0, at which point the agent dies and is cleared off the grid. Agents get rewards for attacking and killing agents on the opposing team.

Figure 9.23. Screenshot from the MAgent multiplayer Gridworld game with two opposing teams of Gridworld agents. The objective is for each team to kill the other.